

Digital Substation Process Bus

Merging Units, Sampled Values, GOOSE, Time Synchronization, and Bay-Level Engineering

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Executive abstract

This paper explains how process bus should be engineered at bay level, including non-conventional instrument transformer interfaces, sampled values design, GOOSE-based trip distribution, time synchronization, redundancy, testing philosophy, retrofit constraints, and typical engineering checks used to validate latency, bandwidth, and timing performance.

Document basis

Document type	Technical white paper
Primary basis	IEC 61850-9-2, IEC 61850-8-1, IEC 61869, IEC 62439-3, IEEE 1588, IEC 61850-6
Intended audience	Utilities, EPC contractors, industrial asset owners, and engineering teams
Publication note	Prepared in formal issuer style to demonstrate CESC engineering capability in studies, protection, digital integration, retrofit, and asset management.

Applicable references

Reference	Application in this paper
IEC 61850-9-2	Sampled values process-bus communication basis.
IEC 61850-8-1	MMS and Ethernet mapping for station-bus data exchange.
IEC 61869	Instrument transformer and low-power measurement interface context.
IEC 62439-3	Redundancy methods such as PRP and HSR.
IEEE 1588	Precision time synchronization context.
IEC 62271-200	MV switchgear context and station implementation environment.
IEC 61850-6	SCL engineering, interoperability documentation, and configuration control basis.

Purpose and scope

Process bus moves measurement and trip-related information from extensive copper circuits into a structured digital environment. The engineering challenge is to preserve protection speed, dependability, and maintainability while reducing marshalling complexity and enabling scalable station design.

This paper focuses on the practical engineering of merging units, sampled values, GOOSE transfer, redundant networks, and testing methods.

Typical design review checks include sampled-values loading per bay, end-to-end trip transfer time, time synchronization accuracy, dataset discipline, and failover behavior during network disturbance or maintenance isolation.

Engineering architecture

At process level, primary quantities are sampled in or near the yard and published over Ethernet using sampled values. Trip, interlocking, status transfer, and breaker failure interfaces may be distributed by GOOSE. The bay and station layers must therefore be designed as one coordinated protection environment rather than separate hardware packages.

The architecture should define publisher-subscriber ownership, failover behavior, time quality, cyber boundary, maintenance isolation, and fallback philosophy.

A practical bay design should also define whether the merging unit is stand-alone or integrated inside a protection IED, how breaker position and trip circuits are acquired, what redundancy method is used, and what fallback state is required during loss of synchronization or loss of one communication path.

Calculation and performance checks

Protection performance must be verified end-to-end. The engineer should confirm that the total chain from primary transducer through merging unit, network, subscriber IED, and output module remains acceptable for the specific protection function.

For bay tripping logic, the total transfer time can be represented as: $t_{\text{total}} = t_{\text{sensor}} + t_{\text{MU}} + t_{\text{network}} + t_{\text{IED}} + t_{\text{output}}$. Each component must be justified in design review and validated during FAT.

The sampled-values traffic load should also be checked against switch bandwidth, multicast design, and redundancy overhead.

Typical engineering checks are shown below in the same practical form used during design review and FAT preparation.

Example 1 - End-to-end trip path check: assume $t_{\text{sensor}} = 0.25$ ms, $t_{\text{MU}} = 1.00$ ms, $t_{\text{network}} = 0.80$ ms, $t_{\text{IED}} = 2.10$ ms, and $t_{\text{output}} = 0.50$ ms. Then $t_{\text{total}} = 0.25 + 1.00 + 0.80 + 2.10 + 0.50 = 4.65$ ms. This is generally acceptable for high-speed bay tripping and shows that process-bus transfer itself is only one part of the overall protection operating time.

Example 2 - Sampled-values stream loading: for 50 Hz systems using 80 samples per cycle, the publishing rate is $50 \times 80 = 4,000$ frames/s. If one sampled-values frame occupies approximately 154 bytes on the wire, the traffic per stream is $4,000 \times 154 \times 8 = 4,928,000$ bit/s, or about 4.93 Mb/s. Under PRP, the same stream is transmitted on both LANs, so the effective network reservation becomes about 9.86 Mb/s per stream. This confirms why bay count, multicast filtering, and switch backplane margin must be checked early.

Example 3 - Time synchronization budget: assume grandmaster accuracy contribution = 0.20 microseconds, network residence and asymmetry contribution = 0.35 microseconds, and subscriber recovery/jitter contribution = 0.15 microseconds. The total estimated synchronization error is $0.20 + 0.35 + 0.15 = 0.70$ microseconds. This is within a 1 microsecond engineering target and supports correct time alignment for sampled values and sequence-of-events recording.

Example 4 - Bay-level network loading check: if one bay publishes 2 sampled-values streams and 40 GOOSE/status messages with low steady-state traffic, a conservative design may reserve about $2 \times 9.86 = 19.72$ Mb/s for redundant sampled-values traffic and keep additional margin for burst events, engineering access, and supervision traffic. On a 100 Mb/s port this is workable for a disciplined bay design, but aggregation to multiple bays must be checked at switch uplinks and station backbone level.

These example values are not universal settings. They are typical engineering calculations used to verify that the selected process-bus architecture is technically defensible before procurement, FAT, SAT, and brownfield cutover.

Implementation controls

A process-bus project needs a disciplined signal and dataset philosophy. Dataset growth without engineering control can create bandwidth waste, difficult troubleshooting, and subscription errors.

Testing should include time synchronization loss, network path interruption, subscriber failover, sampled-values quality flags, and restoration behavior after switch replacement or configuration reload.

Configuration control should include sampled-values identifiers, VLAN and priority mapping, GOOSE dataset freeze, SCL revision management, and defined regression tests after firmware or network changes.

Retrofit strategy

The most practical retrofit path is often a hybrid station: conventional CT/VT and hardwired trips retained where necessary, with process-bus pilots applied first to selected bays or new extensions.

This allows operational staff to gain confidence in digital workflows without forcing a single-step transformation across the whole station.

CESC engineering contribution

CESC can package process-bus work as a controlled engineering deliverable: architecture definition, signal ownership schedule, network concept, time synchronization design, testing procedures, and staged brownfield deployment planning.

In practice, this means CESC can convert abstract IEC requirements into auditable design criteria, calculation sheets, network loading checks, FAT/SAT procedures, and brownfield migration packages that utilities and EPC clients can implement with confidence.

Conclusion

These website papers are structured to demonstrate how CESC approaches engineering subjects at system level, not only at equipment catalogue level. The intended message to clients is that CESC can define studies, design criteria, scheme architecture, implementation strategy, testing approach, and brownfield execution method across protection, control, digital integration, retrofit, and asset management scopes. The added value from CESC is the ability to translate standards and vendor data into practical engineering decisions, calculation-backed design checks, and controlled implementation methodology that reduce risk for utilities, EPC contractors, and industrial asset owners.